Companion LVI: Galaxy–Galaxy Lensing in the Relaxation-Driven Cyclic Cosmology

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Abstract

Galaxy–galaxy lensing (GGL) probes the relationship between matter and the gravitational potential by measuring the distortion of background galaxies around foreground lenses. In the standard cosmological model, the GGL signal is determined by the projected matter density profile and the growth of structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the matter distribution, the potential depth, and the tangential shear profile in a characteristic way. This Companion develops a continuous description of galaxy–galaxy lensing in RDCC, deriving how the relaxation scale influences the excess surface density, the tangential shear, and the observational signatures in weak-lensing surveys. The resulting framework provides a direct probe of the RDCC gravitational field on halo and large-scale structure scales.

1 Introduction

Galaxy–galaxy lensing measures the distortion of background galaxies due to the gravitational potential of foreground galaxies or halos. The observable is the tangential shear, which is directly related to the excess surface density (ESD). GGL provides a powerful probe of the matter distribution, halo profiles, and the growth of structure.

In the standard cosmological model, the GGL signal is determined by the projected matter density profile and the linear growth rate. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the matter distribution, the potential depth, and the tangential shear profile in a scale-dependent manner, producing a distinctive signature in galaxy–galaxy lensing.

2 The RDCC Matter Density Profile

The matter density contrast evolves as

$$\delta_{m,\mathrm{RDCC}}(k,z) = \delta_m(k,z)\left[1 - \chi_\delta \left(\frac{k}{k_{\mathrm{rel}}}\right)^{\alpha}\right]. \tag{1}$$

The projected surface density becomes

$$\Sigma_{\mathrm{RDCC}}(R) = \Sigma(R)\left[1 - \chi_\Sigma \left(\frac{R_{\mathrm{rel}}}{R}\right)^{\alpha}\right]. \tag{2}$$

This modification suppresses small-scale density and enhances large-scale coherence.

3 Excess Surface Density in RDCC

The excess surface density is defined as

$$\Delta\Sigma(R) = \bar{\Sigma}(< R) - \Sigma(R). \tag{3}$$

In RDCC, this becomes

$$\Delta\Sigma_{\mathrm{RDCC}}(R) = \Delta\Sigma(R) \left[1 - \chi_{\Delta\Sigma} \left(\frac{R_{\mathrm{rel}}}{R}\right)^{\alpha}\right]. \tag{4}$$

This modification reflects the scale-dependent evolution of the gravitational potential.

4 Tangential Shear in RDCC

The tangential shear is given by

$$\gamma_t(R) = \frac{\Delta\Sigma(R)}{\Sigma_{\mathrm{crit}}}. \tag{5}$$

In RDCC, this becomes

$$\gamma_{t,\mathrm{RDCC}}(R) = \gamma_t(R) \left[1 - \chi_{\gamma} \left(\frac{R_{\mathrm{rel}}}{R}\right)^{\alpha}\right]. \tag{6}$$

This modification suppresses small-scale shear and enhances large-scale coherence.

5 Large-Scale Coherence and RDCC Signatures

The relaxation mechanism enhances the coherence of large-scale modes. The GGL signal therefore exhibits:

- enhanced shear at large radii,
- suppressed shear at small radii,
- a characteristic turnover near R_{rel},
- a modified redshift dependence of the ESD amplitude.

These signatures provide a direct observational probe of the RDCC gravitational field.

6 Conclusion

Galaxy–galaxy lensing in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential and the matter distribution. The relaxation mechanism modifies the matter density profile, the excess surface density, and the tangential shear in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of halo structure, matter evolution, and the interplay between light and gravity in RDCC.

References

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